Business Requirements Document: Customer Churn Analysis

Telecom customer churn (IBM Telco dataset). Author: Zubair Hashmat. Date: 11-11-2026. Version 1.0.

1. Business Problem

A telecommunications provider is losing customers and does not have a clear picture of **which customers leave and why**. Acquiring a new customer typically costs more than keeping an existing one, so understanding churn is a priority for the retention team. Today there is no single view of who is churning, what they have in common, what they say about their reasons for leaving, or how much revenue is at stake.

2. Objectives

1. Describe the customers who churned compared with those who stayed.
2. Identify the reasons customers give for leaving and rank them.
3. Identify the account, service and demographic patterns most associated with churn.
4. Quantify the revenue and customer lifetime value lost to churn.
5. Give the retention team a dashboard they can use to find high-risk segments.

3. Business Questions

#	Question	Type
1	What is the overall churn rate, and how much monthly revenue does it represent?	Headline KPI
2	Who churns: how does churn vary by age group, gender, family status and location?	Customer profile
3	Does churn depend on contract type, tenure, payment method or billing type?	Account drivers
4	Which services (internet type, security, tech support, streaming and so on) are linked with higher or lower churn?	Service drivers

#	Question	Type
5	How does churn change with monthly charges?	Pricing
6	What reasons do churned customers give, and which categories dominate?	The "why"
7	Which customer segments combine high churn with high customer value (CLTV)?	Prioritisation

4. Stakeholders and Audience

Stakeholder	Interest
Retention manager (primary)	Which segments to target, and what to fix
Marketing / customer success	Messaging and offers for at-risk segments
Product / pricing	Service and price issues behind churn
Executive sponsor	Headline churn rate and revenue impact

The dashboard should be readable by a non-technical manager in under two minutes.

5. Scope

In scope

- Descriptive analysis of the IBM Telco customer churn dataset.
- Data profiling in Excel, then cleaning and analysis in **PostgreSQL**.
- An interactive dashboard in **Power BI Desktop**, connected to PostgreSQL (five report pages).
- Documentation: this BRD, SQL scripts, a data cleaning log and a README.

Out of scope

- Building or deploying a predictive churn model.
- Real-time or automated data refresh.
- Time-series trend analysis (the data is a single-quarter snapshot).

- Recommending specific prices or discount amounts.

6. Data

Item	Detail
Source	Telco Customer Churn: IBM dataset, Kaggle (https://www.kaggle.com/datasets/yeanzc/telco-customer-churn-ibm-dataset)
Content	33 variables covering demographic, geographic, service and account information, plus churn indicators and customer lifetime metrics
Grain	One row per customer
Target	Churn label (customer left or stayed during the quarter)
Row count	7,043 customers (1,869 churned = 26.54%; 5,174 retained)
Column names and types	33 columns, confirmed after download: customer ID, location, demographics, services, account and billing, churn label and value, churn score, CLTV and churn reason. Constant or redundant columns (Count, Country, State, Lat Long, Churn Value) are dropped in the clean table

Fields that need special handling

- **Churn Reason:** populated only for churned customers (1,869 filled; the 5,174 blanks are exactly the retained customers, so they are structural, not missing data), so it describes complaints and cannot be compared with customers who stayed.
- **Churn Score:** an output of IBM's own predictive model. It must not be reported as a driver of churn (circular). Use it for context only.
- **CLTV:** a predicted customer lifetime value. Use it to prioritise segments, not to explain churn.
- **Total Charges:** 11 blank values, all for new customers with tenure 0 who have not yet been billed (all retained). Keep as NULL; do not set to 0.

- **No internet service / No phone service:** a separate category that overlaps with No. Keep the rows; compare add-on services among internet customers only.

7. Key Performance Indicators

KPI	Definition
Churn rate	Churned customers divided by total customers
Churned monthly revenue	Sum of monthly charges for churned customers
Revenue churn rate	Churned monthly revenue divided by total monthly revenue
Average tenure of churners	Mean tenure in months for churned customers, compared with retained customers
Top churn reasons	Share of churned customers by reason category
CLTV at risk	Sum of CLTV for churned customers (and for segments with above-average churn)

Results from the completed analysis

KPI	Result
Churn rate	26.54% (1,869 of 7,043 customers)
Churned monthly revenue	139,130.85 (30.5% of total monthly revenue)
Average tenure of churners	18.0 months (retained customers: 37.6)
Average monthly charge	74.44 for churned vs 61.27 for retained customers
Top churn reason group	Competitor, 33.2% of churners (top single reason: attitude of support person, 10.3%)
CLTV at risk	7,755,256 (predicted value, used for ranking only)
Highest-impact segment	Month-to-month, fiber optic, electronic check: 18.6% of customers, 42.2% of churners, about 49% of monthly revenue lost

8. Functional Requirements: Dashboard

1. **Overview page:** KPI cards (customers, churn rate, churned revenue, average tenure), plus churn rate by contract type and tenure band, and a short insight panel.
2. **Customer profile page:** churn by age group, partner and dependents status, gender and city (cities with 50 or more customers).
3. **Drivers page:** contract by tenure and internet service by monthly charges matrices, payment method, and add-on services (internet customers only).
4. **Reasons page:** ranked churn reason groups and individual reasons, and the reason mix by contract type, with slicers.
5. **Filters (slicers):** slicers are used on the Reasons page only (contract type, internet service, tenure band, age group), so chart titles on the other pages can state fixed findings.
6. **Design:** consistent colours, clear titles that state the finding, labelled axes and units, and a short note on data definitions.

Priority segments page (added in version 1.1): the ten customer segments (contract, internet service and payment method; 100 or more customers each) with the most churned customers, showing churn rate, monthly revenue lost and CLTV lost, plus cards for the highest-impact segment.

9. Approach and Tools

Step	Tool	Output
1. Profile the data	Excel (working copy; raw file untouched)	Issues log, early pivot-table hypotheses
2. Load and clean	PostgreSQL	Raw table, cleaned view or table, cleaning log
3. Analyse	PostgreSQL	Query scripts answering each business question
4. Visualise	Power BI Desktop (Windows)	Five-page dashboard (Overview, Customer profile, Drivers, Reasons, Priority segments)
5. Document	GitHub	README, SQL scripts, dashboard screenshots

10. Assumptions and Risks

Assumption or risk	Handling
Churn label reflects customers who left during a single quarter, so this is a snapshot, not a trend	State clearly in the dashboard and README
Churn is the minority outcome (26.54% of customers: 1,869 of 7,043)	Report both counts and percentages; avoid headline claims from small segments
Churn Reason covers churned customers only	Present it as a ranking of stated reasons, not as a cause
Churn Score is a model output	Exclude from driver analysis; mention it as context only
Findings are correlations, not causes	Use careful wording ("associated with")
Numeric fields may be stored as text or contain blanks	Checked in Excel and SQL: 11 blank Total Charges (new customers, tenure 0) kept as NULL; all fixes logged in SQL

Assumption or risk	Handling
The dataset is fictional sample data	Note that results illustrate method, not a real company
Segments overlap (for example month-to-month, fiber and electronic check)	Use cross-tabs and segment views; never add effects together
Small groups (for example two-year customers in their first 24 months: 68 and 90 customers)	Show group sizes next to rates; avoid strong claims from small cells
Churn reason groups are the analyst's own categorisation of 20 reasons	Document the mapping in SQL and state it in the README and dashboard

11. Deliverables

1. This BRD (version 1.1).
2. Excel profiling notes and issues log.
3. SQL scripts: 01_setup_and_clean.sql and 02_analysis.sql (load, validate, clean, views and analysis queries).
4. Power BI dashboard file (.pbix), PDF export and screenshots of all five pages.
5. README with executive summary, findings, cleaning log and limitations.

12. Success Criteria

- Each business question in section 3 is answered with a number or chart.
- Every figure on the dashboard can be traced to a SQL query.
- The cleaning steps are documented and reproducible.
- A non-technical reader can identify the top churn segments and top reasons within two minutes.

13. Timeline (original estimate)

Phase	Estimate
BRD and Excel profiling	1 to 2 days

Phase	Estimate
PostgreSQL load, cleaning and analysis	3 to 4 days
Power BI dashboard	3 to 4 days
README and polish	1 to 2 days

14. Approval

Role	Name	Date
Author	[Your name]	[Date]
Reviewer (optional: mentor or peer)	[Name]	[Date]